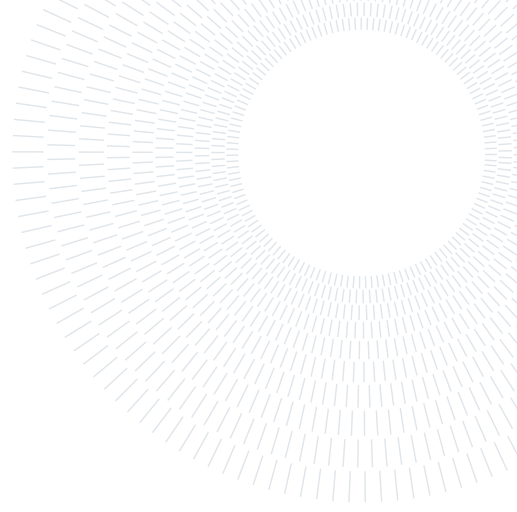




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054702 - AUTOMATION AND CONTROL LABORATORY

LABORATORY REPORT FOR THE SETUP: SEESAW

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1. Introduction

2. Setups Descriptions

2.1. Cart Subsystem

The cart is a 1-DOF linear system. Parameters are derived from the hardware datasheet and validated experimentally in Section 3.2.

The actual weight of the cart seen by the motor is

$$M_e = M_c + \frac{J_m \cdot K_g^2}{r_{mp}^2} \quad (1)$$

Defining the force acting on the cart as

$$F_c(t) = \alpha \cdot v(t) - B_{emf} \cdot \dot{x}_c(t) \quad (2)$$

where

$$\alpha = \frac{k_t K_g \eta_g \eta_m}{r_{mp}^2 R_m} \quad B_{emf} = \frac{k_t k_m K_g^2 \eta_g}{r_{mp}^2 R_m}$$

The kinetic energy takes the form

$$T = \frac{1}{2} M_e \dot{x}_c^2 \quad (3)$$

The equation of motion of the system is:

$$\begin{aligned}
\frac{\partial^2 T}{\partial t \partial \dot{x}_c} - \frac{\partial T}{\partial x_c} &= -B_{eq,c} \cdot \dot{x}_c + F_c \\
\Leftrightarrow M_e \cdot \ddot{x}_c &= -B_{eq,c} \cdot \dot{x}_c + \alpha \cdot v - B_{emf} \cdot \dot{x}_c \\
\Leftrightarrow M_e \cdot \ddot{x}_c(t) + (B_{eq,c} + B_{emf}) \cdot \dot{x}_c(t) &= \alpha \cdot v(t)
\end{aligned} \tag{4}$$

2.2. Seesaw and Cart System

The coupled 2-DOF system consists of cart position x_c and seesaw inclination θ . The Lagrangian $L = T - V$ is used to derive the equations of motion.

$$T = \frac{1}{2}M_e\dot{x}_c^2 + \frac{1}{2}(J_{sw} + M_e D_T^2)\dot{\theta}^2 + \frac{1}{2}M_e x_c^2 \dot{\theta}^2 - M_e D_T \dot{x}_c \dot{\theta} \tag{5}$$

$$V = gM_e x_c \sin(\theta) + g(M_e D_T + M_{sw} D_C) \cos(\theta) \tag{6}$$

Using the Euler-Lagrange relation:

$$\begin{cases} \frac{\partial^2 L}{\partial t \partial \dot{x}_c} - \frac{\partial L}{\partial x_c} = -B_{eq,c} \dot{x}_c + F_c \\ \frac{\partial^2 L}{\partial t \partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = -B_{sw} \dot{\theta} \end{cases} \tag{7}$$

The final equations of motion for the complete system are:

$$\begin{cases} M_e \ddot{x}_c - M_e D_T \ddot{\theta} - M_e x_c \dot{\theta}^2 + gM_e \sin(\theta) = -B_{eq,c} \dot{x}_c + F_c \\ M_e \ddot{\theta} x_c^2 + 2M_e \dot{x}_c \dot{\theta} + gM_e \cos(\theta) x_c - M_e D_T \ddot{x}_c + \\ (J_{sw} + M_e D_T^2) \ddot{\theta} - g(M_e D_T - M_{sw} D_C) \sin(\theta) = -B_{sw} \dot{\theta} \end{cases} \tag{8}$$

3. First experience: Cart position control

3.1. Transfer Function Definition

Based on the equation of motion described at Eq. 4, we define the transfer function of the system using the Laplace Transform.

$$[M_e \cdot s^2 + (B_{eq,c} + B_{emf}) \cdot s] \cdot X_c(s) = \alpha \cdot V(s) \tag{9}$$

$$\begin{aligned} G(s) &= \frac{X(s)}{V(s)} = \frac{\alpha}{M_e \cdot s^2 + B_{tot} \cdot s} \\ &= \frac{1}{s} \cdot \frac{(\alpha/B_{tot})}{(M_e/B_{tot}) \cdot s + 1} \end{aligned} \tag{10}$$

3.2. System Identification and Tuning

Parameters $B_{eq,c}$, η_g , and η_m are identified through experimental data fitting. Nominal values from Table 1 serve as initial estimates.

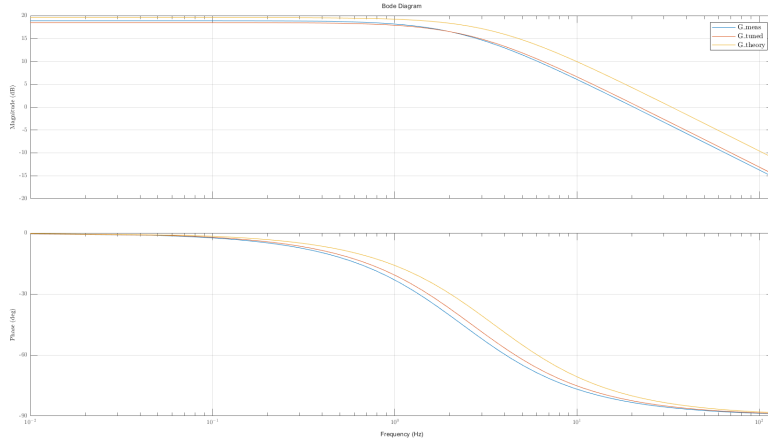


Figure 1: Caption

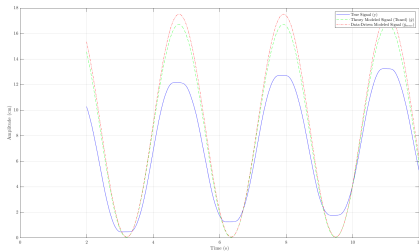
Friction and Efficiency Tuning Viscous coefficient $B_{eq,c}$ is identified via sine-sweep excitation and mean-square error fitting. Efficiencies η_g and η_m are tuned using step-response time constants and DC gain. A grid search identifies values that account for unmodelled Coulomb friction and cable drag.

$$\tau = \frac{M_e}{B_{tot}(\eta_g)}, \quad K = \frac{\alpha(\eta_g, \eta_m)}{B_{tot}(\eta_g)}$$

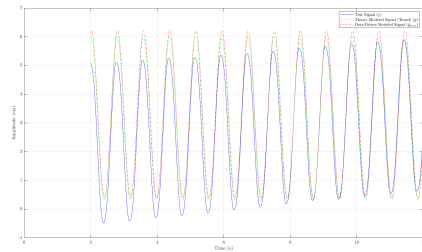
	η_g	η_m	mean square error
datasheet base value	0.9	0.69	$1.11 \cdot 10^2$
datasheet variation	0.81	0.65	$8.08 \cdot 10^1$
extra variation	0.72	0.5865	$4.98 \cdot 10^1$

Table 1: Efficiencies and associated MSE depending on the setup.

The identified model parameters that provide the best fit to the measured step response are $B_{eq,c} = 3.87$, $\eta_g = 0.6$, and $\eta_m = 0.69$ (see Fig. 1).



(a) One Polimi logo.



(b) Another one Polimi logo.

Figure 2: Caption of the Figure.

final values $\eta_g = 0.72$ [-], $\eta_m = 0.5865$ [-]

3.3. Model Validation

Time validation

Frequency Validation

3.4. Control of the position

using PID control

4. Second experience: Body Stabilization in horizontal position

4.1. Transfer Functions Definition

Assumptions for linearization :

1. Small angles ($\theta \simeq 0$) : $\sin(\theta) = \theta$ and $\cos(\theta) = 1$
2. Ignoring high-order velocity terms : $\dot{\theta}^2, x_c^2, \dot{x}_c \dot{\theta} \simeq 0$

After linearizing and replacing F_c , the system is:

$$\begin{cases} M_e \ddot{x}_c - M_e D_T \ddot{\theta} + g M_e \theta + B_c \dot{x}_c + B_{emf} \dot{x}_c = \alpha V_{DC} \\ -M_e D_T \ddot{x}_c + g M_e x_c + \\ (J_{sw} + M_e D_T^2) \ddot{\theta} + B_{sw} \dot{\theta} - g(M_e D_T - M_{sw} D_C) \theta = 0 \end{cases} \quad (11)$$

For which the Laplace transform gives, considering initial states of (0,0):

$$\begin{cases} (M_e s^2 + (B_{eq,c} + B_{emf})s) \cdot X(s) + \\ (-M_e D_T s^2 + g M_e s) \cdot \Theta(s) = \alpha \cdot V_{DC}(s) \\ (-M_e D_T s^2 + g M_e) \cdot X(s) + \\ ((J_{sw} + M_e D_T) s^2 + B_{sw} s - g(M_e D_T + M_{sw} D_C)) \cdot \Theta(s) = 0 \end{cases} \quad (12)$$

Matrix form :

$$\begin{bmatrix} M_e s^2 + (B_{eq,c} + B_{emf})s & -M_e D_T s^2 + g M_e s \\ -M_e D_T s^2 + g M_e & (J_{sw} + M_e D_T) s^2 + B_{sw} s - g(M_e D_T + M_{sw} D_C) \end{bmatrix} \begin{bmatrix} X(s) \\ \Theta(s) \end{bmatrix} = \begin{bmatrix} \alpha \\ 0 \end{bmatrix} V_{DC}(s) \quad (13)$$

$$\begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} \begin{bmatrix} X(s) \\ \Theta(s) \end{bmatrix} = \begin{bmatrix} \alpha \\ 0 \end{bmatrix} V_{DC}(s) \quad (14)$$

using Cramer's rule we can define $G(s)$ and $H(s)$ the transfer functions from $V_{DC}(s)$ to respectively $X(s)$ and $\Theta(s)$.

$$\begin{aligned} G(s) &= \frac{X(s)}{V_{DC}(s)} = \alpha \frac{P_{22}}{P_{11} \cdot P_{22} - P_{12} \cdot P_{21}} \\ &= \frac{numG}{den} \\ H(s) &= \frac{\Theta(s)}{V_{DC}(s)} = \alpha \frac{-P_{21}}{P_{11} \cdot P_{22} - P_{12} \cdot P_{21}} \\ &= \frac{numH}{den} \end{aligned} \quad (15)$$

Both share the same 4 poles : -23 , -3.31 , 0 , 3

Having the 4th pole positive makes the system unstable and this requires the control actions developed in the next sections.

4.2. State-Space Representation

Solving the equations of motions for the second order derivative:

$$\begin{bmatrix} M_e & -M_e D_T \\ -M_e D_T & J_{sw} + M_e D_T^2 \end{bmatrix} \begin{bmatrix} \ddot{x}_c \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} 0 & -gM_e & -B_{tot} & 0 \\ -gM_e & g(M_e D_T + M_{sw} D_C) & 0 & -B_{sw} \end{bmatrix} \begin{bmatrix} x_c \\ \theta \\ \dot{x}_c \\ \dot{\theta} \end{bmatrix} + \begin{bmatrix} \alpha \\ 0 \end{bmatrix} V \quad (16)$$

$$\begin{aligned} \begin{bmatrix} \ddot{x}_c \\ \ddot{\theta} \end{bmatrix} &= \frac{1}{J_{sw}} \begin{bmatrix} -gM_e D_T & g(-J_{sw} + 2gM_e D_T^2 + M_{sw} D_T D_C) & -B_{tot}(\frac{J_{sw}}{M_e} + D_T^2) & -B_{sw} D_T \\ -gM_e & g(2M_e D_T + M_{sw} D_C) & -B_{tot} D_T & -B_{sw} \end{bmatrix} \begin{bmatrix} x_c \\ \theta \\ \dot{x}_c \\ \dot{\theta} \end{bmatrix} \\ &+ \frac{\alpha}{J_{sw}} \begin{bmatrix} \frac{J_{sw}}{M_e} + D_T^2 \\ D_T \end{bmatrix} V \end{aligned} \quad (17)$$

We can extend this system to include $\dot{x}_c$ and $\dot{\theta}$:

$$\begin{aligned} \begin{bmatrix} \dot{x}_c \\ \dot{\theta} \\ \ddot{x}_c \\ \ddot{\theta} \end{bmatrix} &= \frac{1}{J_{sw}} \begin{bmatrix} 0 & 0 & J_{sw} & 0 \\ 0 & 0 & 0 & J_{sw} \\ -gM_e D_T & g(-J_{sw} + 2gM_e D_T^2 + M_{sw} D_T D_C) & -B_{tot}(\frac{J_{sw}}{M_e} + D_T^2) & -B_{sw} D_T \\ -gM_e & g(2M_e D_T + M_{sw} D_C) & -B_{tot} D_T & -B_{sw} \end{bmatrix} \begin{bmatrix} x_c \\ \theta \\ \dot{x}_c \\ \dot{\theta} \end{bmatrix} \\ &+ \frac{\alpha}{J_{sw}} \begin{bmatrix} 0 \\ 0 \\ \frac{J_{sw}}{M_e} + D_T^2 \\ D_T \end{bmatrix} V \end{aligned} \quad (18)$$

This gives the state and input matrices as

$$\begin{aligned} A &= \frac{1}{J_{sw}} \begin{bmatrix} 0 & 0 & J_{sw} & 0 \\ 0 & 0 & 0 & J_{sw} \\ -gM_e D_T & g(-J_{sw} + 2gM_e D_T^2 + M_{sw} D_T D_C) & -B_{tot}(\frac{J_{sw}}{M_e} + D_T^2) & -B_{sw} D_T \\ -gM_e & g(2M_e D_T + M_{sw} D_C) & -B_{tot} D_T & -B_{sw} \end{bmatrix} \\ B &= \frac{\alpha}{J_{sw}} \begin{bmatrix} 0 \\ 0 \\ \frac{J_{sw}}{M_e} + D_T^2 \\ D_T \end{bmatrix} \end{aligned}$$

Regarding the output equation, only the positions x_c and the angle θ are observable and the measurements are not dependent on the input voltage, therefore the output and transmission matrices are :

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \quad D = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

4.3. Pole Placement State-Feedback Control

4.3.1 Design Objective

The pole-placement controller is designed for the identified linear model around the horizontal equilibrium,

$$\dot{\mathbf{x}} = A\mathbf{x} + Bu, \quad \mathbf{x} = [x_c \quad \dot{x}_c \quad \theta \quad \dot{\theta}]^T. \quad (19)$$

With full-state feedback,

$$u = -\mathbf{K}\mathbf{x}, \quad (20)$$

the closed-loop matrix is $A - B\mathbf{K}$. If the pair (A, B) is controllable, the eigenvalues of $A - B\mathbf{K}$ can be assigned by pole placement [1]. This is a local, linear-model result: it does not prove global stability of the hardware, and it does not remove friction, backlash, encoder quantisation, rail limits, or voltage saturation. Therefore the controller is validated by checking that the measured hardware response remains bounded and respects the physical limits in the tested operating range.

4.3.2 Pole Selection Rationale

The gain vector $\mathbf{K}$ is computed by assigning the closed-loop poles $\{p_1, p_2, p_3, p_4\}$ of $A - B\mathbf{K}$. The pole locations should be interpreted as a design target for the linearised model. In the real plant the measured response also depends on unmodelled nonlinearities and on the observer used to reconstruct the velocity states.

The complex pair is described by

$$p_{1,2} = -\sigma_{th} \pm j\omega_{d,th}, \quad \zeta_{th} = \frac{\sigma_{th}}{\sqrt{\sigma_{th}^2 + \omega_{d,th}^2}}. \quad (21)$$

For a pure second-order system, $t_s \approx 4/\sigma_{th}$ and $M_p \approx \exp(-\pi\zeta_{th}/\sqrt{1-\zeta_{th}^2})$. These formulas are used only as tuning guidance; the seesaw is a coupled fourth-order plant, so the final performance is checked in simulation and on hardware.

Pole roles. The selected pole pattern is:

- $p_{1,2}$: a complex pair chosen as the dominant closed-loop mode associated with balancing the unstable angular motion;
- p_3 : a real pole placed faster than the dominant pair to avoid a slow cart-velocity transient;
- p_4 : a real pole placed slightly faster than the dominant pair to return the cart without making the voltage demand excessive.

Because the states are coupled, these labels are not independent single-state modes. They describe the intended dominant behaviour, not a diagonal decomposition of the physical system.

4.3.3 Open-loop poles

For the current configuration ($M_c = 0.75$ kg, B_{eq} from frequency-sweep tuning):

$$\lambda_{OL} = \{+2.24, -2.65, -0.45, -37.0\} \text{ rad/s} \quad (\text{approximate}) \quad (22)$$

The pole at $p_{OL} \approx +2.24$ rad/s is the unstable mode of the linearised model. Its time constant is approximately $1/p_{OL} = 0.45$ s, which gives the natural fall time of the seesaw near the equilibrium. The controllability matrix of the identified model has rank 4, so all four linear-model poles are assignable for this parameter set.

This statement is restricted to the linear model; if an operating point or parameter set were not controllable or not stabilisable, the uncontrollable unstable modes could not be moved by state feedback.

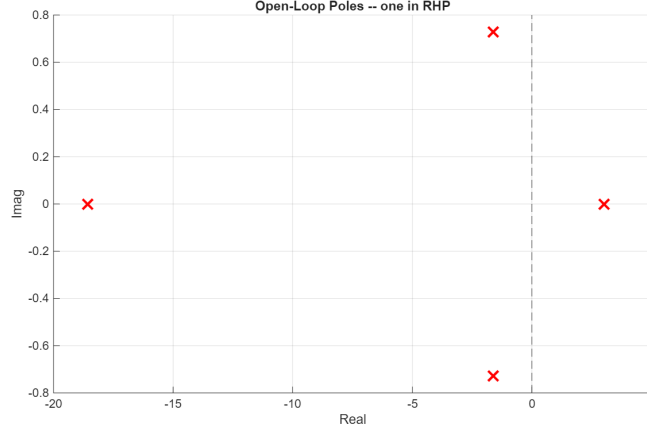


Figure 3: Open-loop pole map. The pole in the right half-plane is the seesaw fall rate p_{OL} and is the design driver for the closed-loop bandwidth.

$p_{1,2}$: **Dominant balancing mode.** Poles $p_{1,2}$ govern the unstable angular dynamics.

- **Decay rate σ_{th} :** Set to 4.5 rad/s, approximately twice the open-loop divergence rate. This makes the linear closed-loop response faster than the natural fall dynamics without requiring unnecessarily high gain.
- **Damping ζ_{th} :** Set to 0.8. The equivalent second-order overshoot is about 1.5%, but the actual overshoot is verified from the complete state-space simulation.

p_3, p_4 : **Non-dominant modes.** These real poles are placed further left than the dominant pair so that they do not introduce slower transients.

- $p_3 = -6.75$ rad/s: a faster real pole used to damp the cart-velocity component of the response.
- $p_4 = -5.4$ rad/s: a real pole used to return the cart toward the centre while keeping the controller less aggressive than a much faster placement.

Final pole set and gain.

$$p_{des} = \{-4.5 \pm j3.37, -6.75, -5.4\} \Rightarrow \mathbf{K} = [147.43 \quad 9.83 \quad -111.00 \quad -36.71] \quad (23)$$

4.3.4 Resulting gain and verification

The gain vector $\mathbf{K}$ is computed by `place(A, B, pdes)`. Numerical values and simulation checks are produced by `pole_placement_design.m`:

$$\mathbf{K} = [147.43 \quad 9.83 \quad -111.00 \quad -36.71]. \quad (24)$$

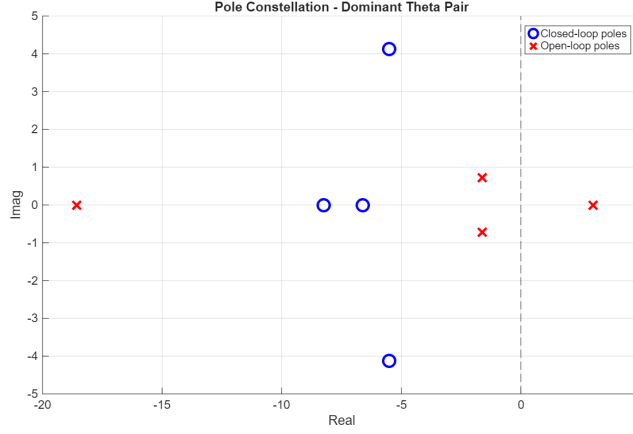


Figure 4: Closed-loop poles of the identified linear model. The complex pair is closest to the imaginary axis and is therefore the intended dominant mode. The real poles are faster, so they should not set the slowest linear transient.

The initial-condition test uses $\theta(0) = 2^\circ$ (Fig. 5). This is a small-angle test inside the linearisation region and close to a realistic manual startup error. Peak values from the linear simulation are:

$$V_m = 3.87 \text{ V}, \quad \theta = 2.55^\circ, \quad x_c = 3.26 \text{ cm}.$$

These values are inside the nominal limits of $\pm 6 \text{ V}$ and $\pm 11.5^\circ$. In the linear simulation, the peak voltage scales with the initial angle, giving $\theta_{\max} = 3.10^\circ$ before V_m reaches saturation. Above this value the actuator saturation changes the closed-loop dynamics, so the assigned poles no longer describe the actual response. The 3.10° value is therefore a conservative bound for the unsaturated linear regime, not a guarantee of recovery for larger disturbances.

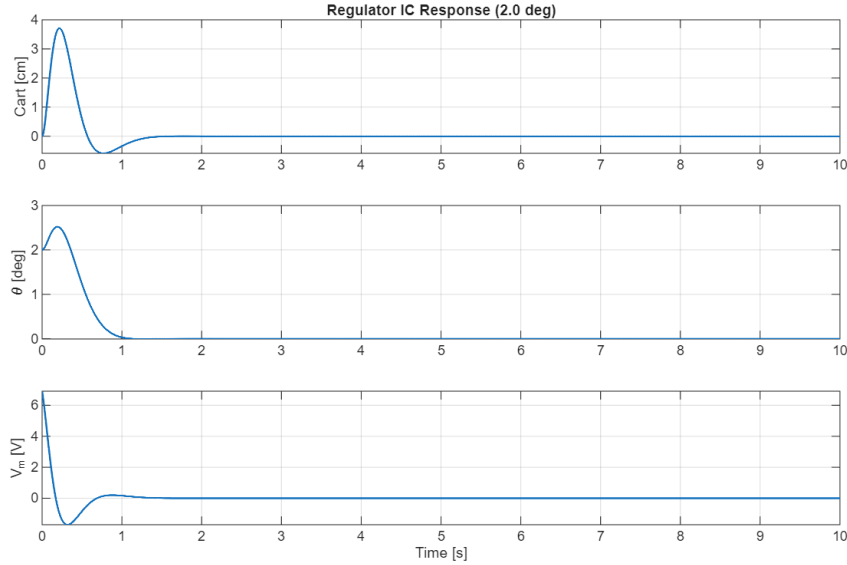


Figure 5: IC response to $\theta(0) = 2^\circ$. Peak $V_m = 3.87 \text{ V}$, peak $\theta = 2.55^\circ$, peak cart = 3.26 cm.

4.3.5 Steady-state bias analysis

An off-centre mass of 100 g induces a constant torque $\tau_{bias} \approx 0.123$ Nm. Pure state feedback does not include a disturbance model or integral state, so a constant torque bias produces a nonzero equilibrium angle θ_{ss} (Fig. 6). On hardware, friction and quantisation can turn this offset into a bounded oscillation around a shifted centre.

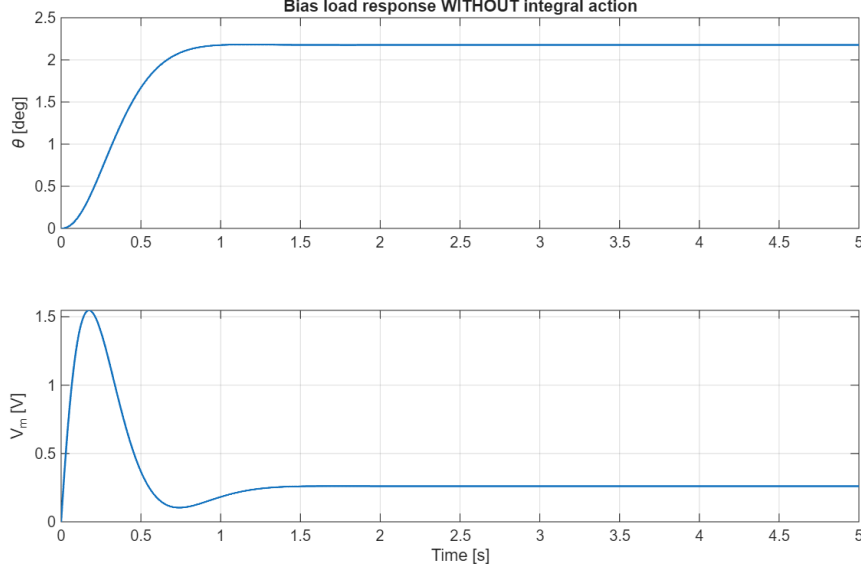


Figure 6: Response to constant torque bias without integral action. Steady-state error θ_{ss} shifts the limit-cycle center.

4.3.6 Integral action

The state is augmented with $\dot{\xi} = \theta$ to eliminate steady-state error:

$$\mathbf{x}_a = [x_c \quad \dot{x}_c \quad \theta \quad \dot{\theta} \quad \xi]^\top. \quad (25)$$

An integrator pole is placed at $p_{int} = -0.5$ rad/s. The integral state rejects constant torque bias in the linear model by driving the average angle back to zero. It should not be interpreted as removing the hardware limit cycle itself; it mainly shifts the centre of the bounded motion back toward $\theta = 0$.

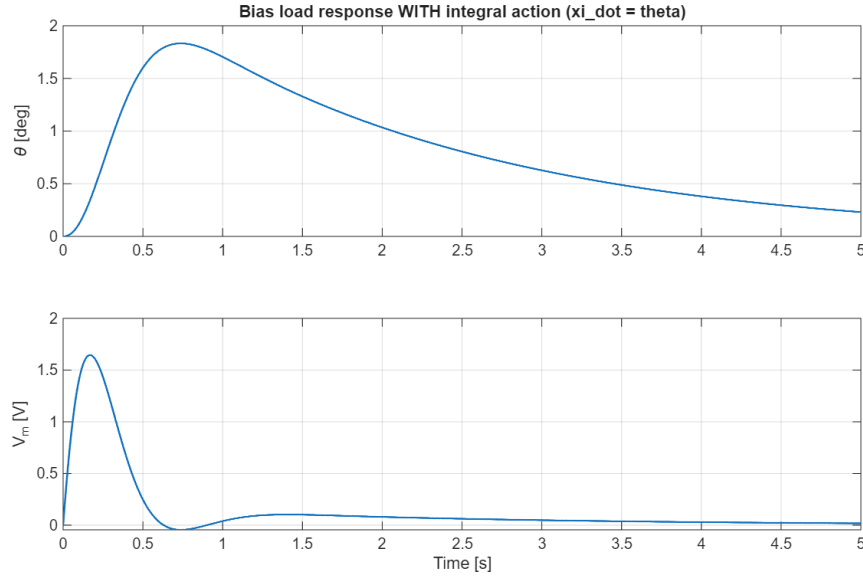


Figure 7: Bias-load response with integral action. Steady-state error is eliminated; oscillation is centered on zero.

4.3.7 Frequency-domain validation

For state feedback there is no unique scalar open-loop transfer function unless the loop-breaking point is specified. The reported frequency plot uses the loop broken at the plant input,

$$L(s) = \mathbf{K}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}. \quad (26)$$

The Bode and Nyquist plots are therefore local linear robustness checks for this specific loop opening. They are useful for comparing gain and phase margins of the implemented design, but they are not a substitute for hardware validation with saturation, quantisation, and observer dynamics present. On the real setup, the corresponding controller frequency response can be estimated from logged data as $C_{hw}(j\omega) \approx U(j\omega)/E(j\omega)$, where E is the controller input signal and U is the commanded motor voltage.

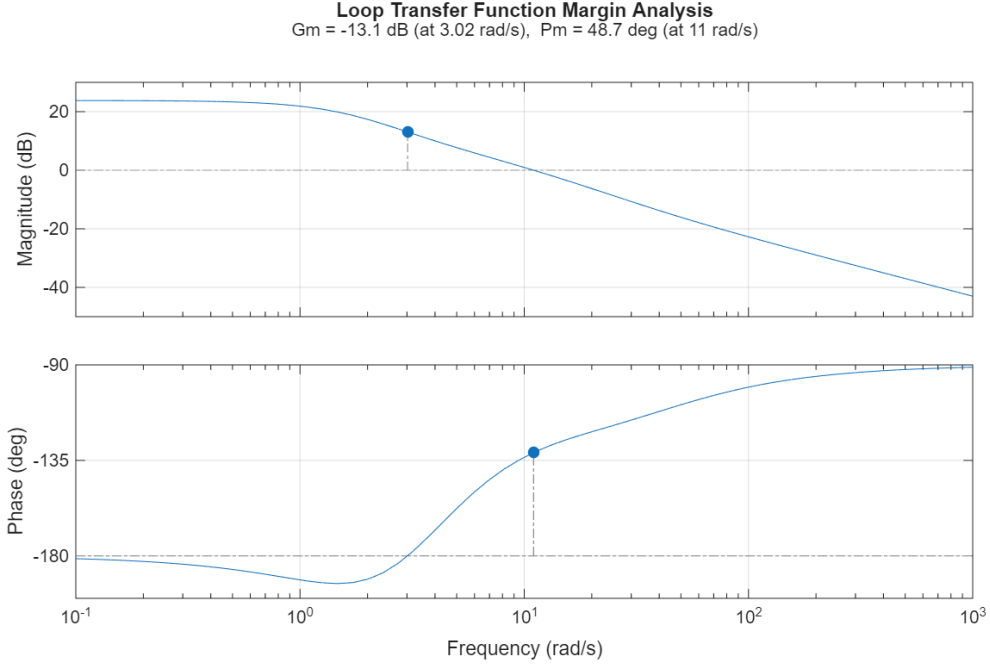


Figure 8: Frequency-domain checks for the linear model with the loop broken at the plant input: Bode plot, Nyquist plot, root locus, and closed-loop pole-zero map. These plots support the linear design but do not replace hardware validation.

4.3.8 Luenberger State Observer

A Luenberger observer is used to estimate the unmeasured states $[\dot{x}_c, \dot{\theta}]$ from position encoders. This replaces numerical differentiation, which introduces significant phase lag.

Theory. The measurement matrix is

$$C_{\text{meas}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad y = C_{\text{meas}} \mathbf{x}. \quad (27)$$

The observer

$$\dot{\hat{\mathbf{x}}} = A\hat{\mathbf{x}} + Bu + L(y - C_{\text{meas}}\hat{\mathbf{x}}) \quad (28)$$

gives an error $e = \mathbf{x} - \hat{\mathbf{x}}$ with dynamics

$$\dot{e} = (A - LC_{\text{meas}})e. \quad (29)$$

L assigns the eigenvalues of $A - LC_{\text{meas}}$ freely whenever (A, C_{meas}) is observable. Closing the loop with $u = -\mathbf{K}\hat{\mathbf{x}}$ gives the block-triangular system

$$\frac{d}{dt} \begin{bmatrix} \mathbf{x} \\ e \end{bmatrix} = \begin{bmatrix} A - B\mathbf{K} & B\mathbf{K} \\ 0 & A - LC_{\text{meas}} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ e \end{bmatrix}, \quad (30)$$

so $\text{eig}(\cdot) = \text{eig}(A - B\mathbf{K}) \cup \text{eig}(A - LC_{\text{meas}})$ – the separation principle. Controller and observer decouple, and $\mathbf{K}$ from §4.3 carries over. The dual identity $\text{eig}(A - LC) = \text{eig}((A^\top - C^\top L^\top)^\top)$ turns the gain problem into pole placement on $(A^\top, C_{\text{meas}}^\top)$, solved with `place` as for $\mathbf{K}$.

Tuning. Let $\sigma_{\text{obs}} = \min_i |\Re(p_{\text{obs},i})|$.

Lower bound – separation. The estimate must settle faster than the controller bandwidth. A standard heuristic

$$|\Re(p_{\text{obs}})| \geq (3-5) |\Re(p_{\text{ctrl}})| \quad (31)$$

separates the observer and controller timescales.

Upper bound – noise. An encoder count perturbs the input by $\delta V_m = \mathbf{KL} \delta y$. With quanta

$$q_{x_c} = K_{ec} \approx 2.28 \times 10^{-5} \text{ m}, \quad q_\theta = K_{E,SW}/K_{gs} \approx 5.0 \times 10^{-4} \text{ rad}, \quad (32)$$

the per-count voltage chatter has a budget $|\mathbf{KL}|_i q_i \lesssim 0.1 \text{ V}$ on each channel. L inflates with k_{obs} , and so does this product.

Damping. Internal to the error dynamics. Mirror the controller pair, $\zeta = 0.8$.

Sweep. Evaluating the noise gain across the target range:

k_{obs}	$ \mathbf{KL} _1 q_{x_c} \text{ [V/count]}$	$ \mathbf{KL} _2 q_\theta \text{ [V/count]}$	$t_{\text{settle}} \text{ [s]}$
1.50	0.000	2.40	1.02
2.00	0.190	0.055	0.84
2.50	0.171	3.71	0.64
3.00	0.148	6.93	0.54
3.50	0.118	9.56	0.46
4.00	0.547	8.25	0.43

Table 2: Observer-gain sweep. The θ -channel noise gain is negligible at $k_{\text{obs}} = 2.0$ (0.055 V/count), then jumps to 3.71 V/count at $k_{\text{obs}} = 2.5$ – the noise-gain knee. Cold-start at $k_{\text{obs}} = 2.5$ exceeds V_{sat} (8.54 V peak), ruling out higher values.

The range $k_{\text{obs}} \geq 2.5$ is unsuitable: the θ -channel noise gain exceeds the chatter budget and cold-start saturation occurs. We pick $k_{\text{obs}} = 2.0$, just below the noise-gain knee:

$$p_{\text{obs}} = 2.0 \cdot \{-4.5 \pm j3.37, -6.75, -5.4\} = \{-9.0 \pm j6.75, -13.5, -10.8\}. \quad (33)$$

This gives a $2\times$ separation ratio. The slowest observer pole at -9.0 rad/s is $3\times$ the unstable open-loop pole ($+3.03 \text{ rad/s}$), satisfying the functional separation requirement.

Verification. The design script `luenberger_observer_design.m` runs all six checks. Numerical results for the chosen $k_{\text{obs}} = 2.0$:

1. **Observability.** $\text{rank}(\mathcal{O}) = 4$, $\text{cond}(\mathcal{O}) = 3.7 \times 10^2$ – well-conditioned. Both states behind each encoder (velocity behind position, $\dot{\theta}$ behind θ) are recoverable.
2. **Gain.**

$$L = \begin{bmatrix} 23.72 & -12.88 \\ -71.75 & 69.97 \\ 29.71 & -0.20 \\ 214.08 & -35.42 \end{bmatrix}. \quad (34)$$

The dominant entries are in the x_c column for the velocity rows and the θ column for $\dot{\theta}$ – the dynamic coupling drives the correction. The error

dynamics decouple from the controller by construction. Eigenvalue mismatch between the combined 8×8 matrix and the union of controller/observer spectra: 8.7×10^{-14} , i.e. machine precision.

3. **IC error decay.** A realistic initial condition of $\theta(0) = 2^\circ$ is used for validation. With $\hat{\mathbf{x}}(0) = 0$ and $\mathbf{x}(0) = [0, 0, 2^\circ, 0]^\top$, all four error components fall inside their encoder-quantum tolerance by $t = 0.90$ s, consistent with the slowest observer pole at -9.0 rad/s.
4. **Loop in the loop.** The combined system response is plotted in Fig. 10.
5. **Noise injection.**

$$|\mathbf{KL}|_1 q_{x_c} = 0.19 \text{ V/count}, \quad |\mathbf{KL}|_2 q_\theta = 0.055 \text{ V/count}. \quad (35)$$

The θ -channel gain matches the controller-only $V_{chatter} = 0.055$ V/count – the observer introduces essentially zero additional noise amplification in this channel.

Noise and bandwidth trade-off. At $k_{\text{obs}} = 2.0$, the θ -channel noise gain is 0.055 V/count, matching the full-state baseline. The observer bandwidth (≈ 9 rad/s) is three orders of magnitude below the sampling rate (1 kHz), so encoder quantisation noise is not passed directly into the control input. On hardware this must be validated against a differentiated encoder signal and a causally low-pass filtered differentiated signal; the expected advantage of the observer is a smoother velocity estimate without the delay required by an aggressively filtered differentiator.

Startup transient. The observer is validated for a 2.0° startup excursion. Peak transient values are compared in Table 3.

Configuration	peak V_m [V]	peak θ [deg]
Full-state controller (baseline)	3.87	2.55
Observer (cold start, $\hat{\mathbf{x}}(0) = 0$)	6.40	2.98
Observer (realistic initialization)	3.87	2.55

Table 3: Transient peak comparison ($\theta(0) = 2^\circ$). Cold-start voltage exceeds V_{sat} . Realistic initialization is required for hardware deployment.

The natural deployment is to seed the observer with the encoders at the moment the controller starts:

$$\hat{\mathbf{x}}(0) = [x_{c,\text{meas}} \quad 0 \quad \theta_{\text{meas}} \quad 0]^\top, \quad (36)$$

zeroing only the unmeasured velocities. With this initialisation the only nonzero error component is the initial velocity guess error (zero in this scripted IC, small in practice), and the observer is transparent: the closed-loop voltage and angle traces match the full-state controller-alone case. Cold-starting from $\hat{\mathbf{x}}(0) = 0$ slightly exceeds V_{sat} (6.40 V peak), so the realistic initialisation from encoder readings is mandatory for safe startup.

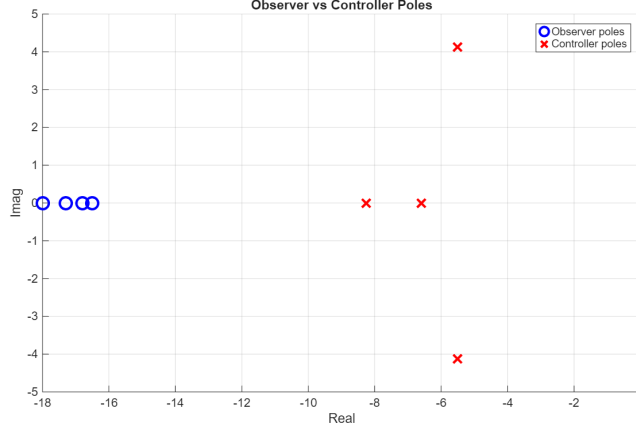


Figure 9: Observer poles (blue) vs controller poles (red). The dominant theta-loop pair sits $2\times$ further left than the controller’s; all observer poles are well into the LHP, with the slowest at -9.0 rad/s.

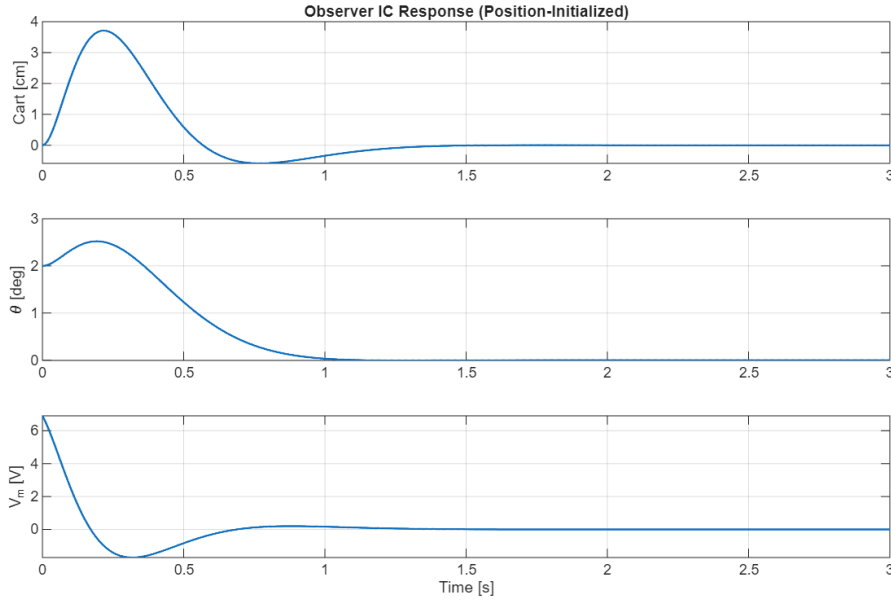


Figure 10: Cold-start IC response with the observer in the loop at $\theta(0) = 2^\circ$. The cart peaks at 3.3 cm, θ overshoots to 3.0° (well inside the $\pm 11.5^\circ$ stops, dashed red), and V_m peaks at 6.40 V, slightly exceeding the saturation envelope (dashed red). With the realistic initialisation (36) the voltage trace drops to the controller-only profile (peak 3.87 V).

Note on the augmented controller. With integral action the controller state is $\mathbf{x}_a = [\mathbf{x}; \xi]$ where $\dot{\xi} = \theta$. Since θ is measured directly, ξ is integrated from the encoder reading, not from $\hat{\theta}$. Only the four plant states are estimated; the observer carries over to the augmented controller without modification.

4.3.9 Implementation and Hardware Results

On hardware, only x_c and θ are measured by encoders. The Luenberger observer of §4.3.8 reconstructs $\dot{x}_c$ and $\dot{\theta}$ from the encoder pair using the linearised plant model, replacing the differentiator + low-pass filter scheme.

Verification Protocol. The observer is verified sequentially to ensure model fidelity:

1. **Open-Loop Tracking:** Verify $\hat{x}_c, \hat{\theta}$ convergence during manual displacement with $V_m = 0$.
2. **Convergence:** Verify software IC convergence ($\hat{\theta}(0) = 15^\circ$) against simulation.
3. **Noise Assessment:** Monitor steady-state motor vibration. Excessive oscillation indicates high observer bandwidth requiring gain reduction.

4.4. AI-Assisted Rapid Prototyping of Robust Controllers

The pole-placement controller and the integral-action extension are both derived from the local linear model. In practice, however, the measured seesaw response showed the same qualitative limitation for several linear designs: instead of converging asymptotically to $\theta = 0$, the hardware settles into a bounded rocking motion. The preceding bias-load study already shows that integral action can recenter the average angle, but it does not remove the hardware limit cycle itself. Since the identified model had proven unreliable outside small local predictions, the next step was not to commit immediately to one controller family, but to rapidly prototype several robust alternatives and compare them on a common nonlinear benchmark.

This exploration was carried out with AI-assisted scripting. The useful role of the AI tool was not to replace control-design judgement, but to accelerate the repetitive parts of the study: generating candidate synthesis scripts, wiring each controller into the same simulation harness, collecting comparable metrics, and quickly discarding controllers that failed the nonlinear constraints. The final decision remains based on the physical metrics: peak-to-peak rocking, voltage effort, voltage slew rate, saturation, and survival under motor-protection limits.

4.4.1 Nonlinear limit-cycle simulation

The robust-controller benchmark uses the identified linear matrices A_{sw}, B_{sw} as the nominal plant, but inserts the main non-ideal hardware effects in the simulation loop. The state-feedback controller receives quantised encoder measurements

$$x_{c,q} = q_x \text{round}\left(\frac{x_c}{q_x}\right), \quad \theta_q = q_\theta \text{round}\left(\frac{\theta}{q_\theta}\right), \quad (37)$$

where $q_x = K_{ec}$ and $q_\theta = K_{E,SW}/K_{gs}$. The motor command is then passed through saturation, deadzone, and asymmetric drive gain:

$$u_s = \text{sat}(u, \pm 6 \text{ V}), \quad u_d = \begin{cases} 0, & |u_s| < 0.12 \text{ V}, \\ u_s - 0.12 \text{ sign}(u_s), & |u_s| \geq 0.12 \text{ V}, \end{cases} \quad (38)$$

$$u_{eff} = \begin{cases} 1.0 u_d, & u_d \geq 0, \\ 0.5 u_d, & u_d < 0. \end{cases} \quad (39)$$

A Coulomb-friction disturbance is added in the cart acceleration channel,

$$d_f = [0 \quad -F_c \text{sign}(\dot{x}_c)/M_c \quad 0 \quad 0]^\top, \quad F_c = 0.18 \text{ N}, \quad (40)$$

and the simulated cart and seesaw are clamped at the physical rail and angle limits. This combination introduces the same mechanism observed on hardware: close to zero angle, the requested voltage often lies inside the deadzone, quantisation prevents infinitesimal feedback corrections, Coulomb friction produces stick-slip motion, and the asymmetric actuator gain biases the two half-cycles. The result is a bounded limit cycle rather than exact convergence.

The baseline pole-placement controller reproduces the large rocking effect in this benchmark. With the same 2° initial release used in the controller tests, the simulated pole-placement steady-state rocking is 3.38° peak-to-peak, matching the order of the hardware observation and motivating the robust-controller comparison.

4.4.2 Controllers compared

The benchmark in `scripts/analysis/robust_controller_comparison.m` evaluates the following controller families on the same nonlinear plant:

- pole-placement state feedback, used as the baseline;
- an LMI-certified state-feedback gain, where high-performing candidates are checked by common quadratic Lyapunov inequalities over uncertainty vertices;
- an H_∞ Riccati state-feedback design;
- an actuator-aware H_∞ design with an augmented actuator-voltage state to penalise aggressive voltage motion;
- super-twisting sliding-mode control (SMC) with a boundary layer;
- a softened scheduled SMC variant;
- a Takagi–Sugeno fuzzy gain scheduler implemented using the Fuzzy Logic Toolbox.

True dynamic `hifsyn` controllers were also attempted after installing the Robust Control Toolbox. Although the resulting dynamic controllers were linearly stabilising, none survived the nonlinear deadzone and constraint validation from the 2° release without falling into saturation or an angle stop. They were therefore excluded from the final surviving-controller table rather than reported as valid.

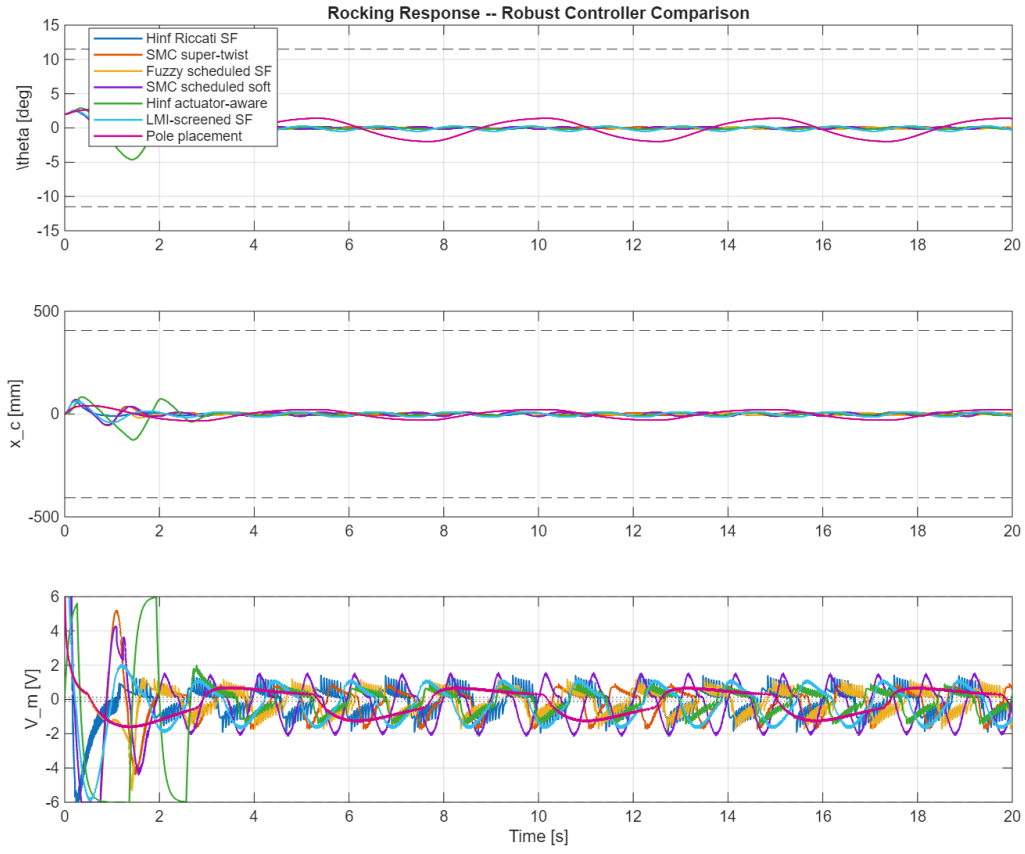


Figure 11: AI-assisted robust-controller comparison on the nonlinear limit-cycle benchmark. All controllers are simulated with the same quantisation, deadzone, asymmetric actuator gain, Coulomb friction, saturation, and stops.

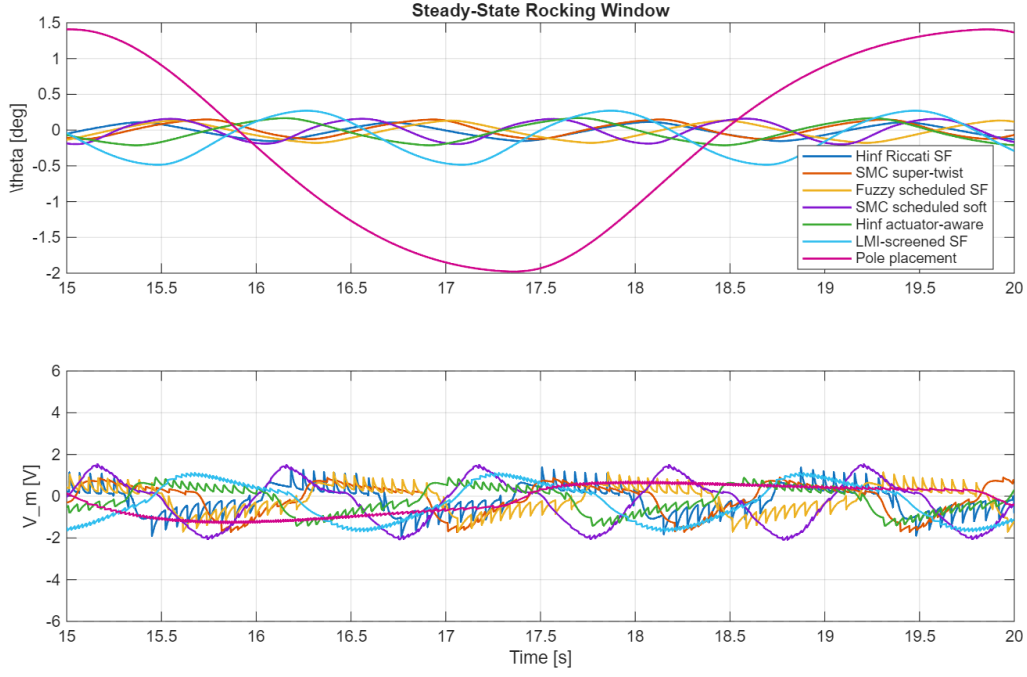


Figure 12: Steady-state rocking window for the robust-controller comparison. The pole-placement controller keeps the plant bounded but leaves a much larger oscillation envelope than the robust candidates.

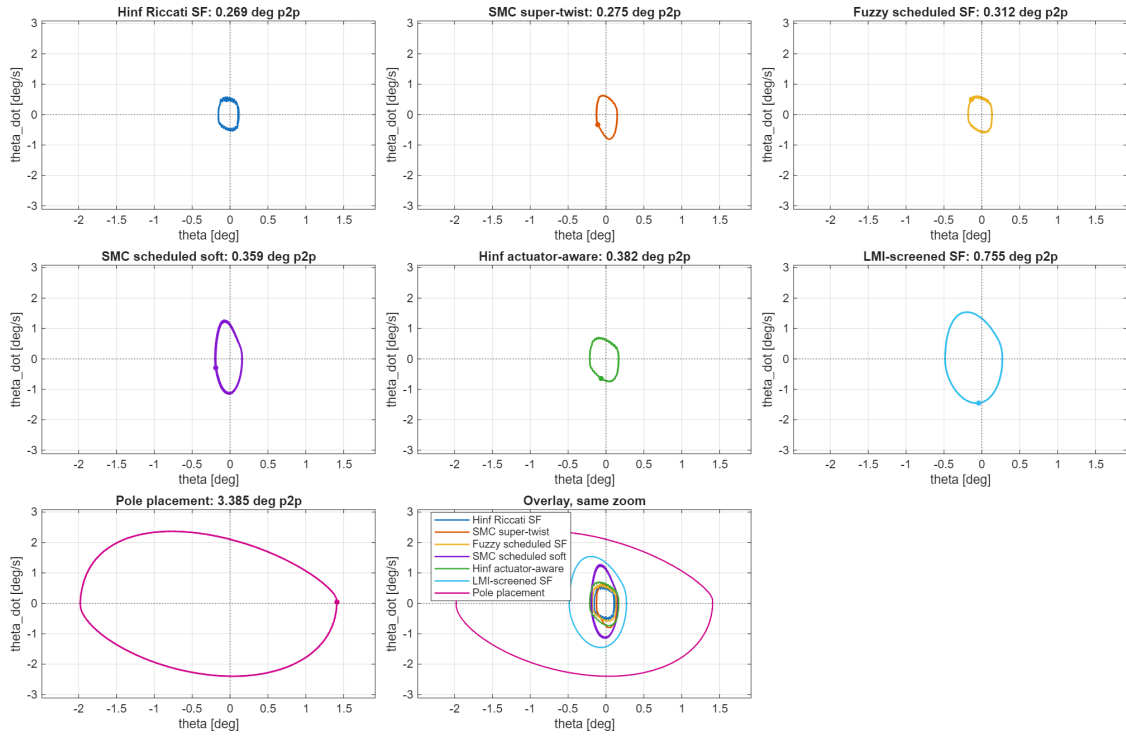


Figure 13: Steady-state phase-plane portraits, θ versus $\dot{\theta}$, for the robust-controller candidates. The plot visualises the remaining limit cycles after the transient has decayed.

4.4.3 Nominal nonlinear benchmark results

Table 4 reports the final comparison metrics over the last five seconds of a 20 s simulation. The primary objective is minimising θ peak-to-peak; control effort and voltage slew are treated as secondary hardware-stress metrics.

Controller	OK	θ_{pp} [deg]	θ_{rms} [deg]	u_{rms} [V]	u_{pk} [V]	peak slew [V/s]
H_∞ Riccati SF	yes	0.269	0.087	0.662	6.000	654.8
SMC super-twist	yes	0.275	0.095	0.759	6.000	270.2
Fuzzy scheduled SF	yes	0.312	0.104	0.638	6.000	338.5
SMC scheduled soft	yes	0.359	0.124	1.127	6.000	609.3
H_∞ actuator-aware	yes	0.382	0.132	0.640	6.000	120.0
LMI-screened SF	yes	0.755	0.280	0.967	6.000	108.9
Pole placement	yes	3.385	1.235	0.726	6.000	153.6

Table 4: Nominal nonlinear robust-controller benchmark. Metrics are computed on the steady-state window $t \geq 15$ s.

The best nominal peak-to-peak result is obtained by the H_∞ Riccati state-feedback controller, closely followed by the super-twisting SMC. However, both achieve this by allowing relatively aggressive voltage changes. The actuator-aware H_∞ controller is more hardware-friendly: it increases the angle envelope to 0.382° peak-to-peak but reduces the peak voltage slew to approximately 120 V/s and avoids saturation in the nominal run. The LMI-screened controller also has a low slew requirement, but at the cost of a larger 0.755° oscillation.

4.4.4 Motor-protection and slew-rate survival

To evaluate weak-hardware operation, the same controllers were tested with an explicit voltage slew-rate limiter. This is not intended as the final design method; it is a stress test showing which controllers remain viable if the motor amplifier must be protected from fast voltage changes. Table 5 gives the lowest slew-rate limit at which each controller still satisfies the simulation constraints.

Controller	lowest surviving limit [V/s]	θ_{pp} [deg]	u_{rms} [V]
Pole placement	10	3.421	0.767
Fuzzy scheduled SF	35	0.341	0.643
LMI-screened SF	50	0.760	0.978
H_∞ Riccati SF	100	0.280	0.616
H_∞ actuator-aware	100	0.382	0.661
SMC super-twist	100	0.273	0.766
SMC scheduled soft	100	0.360	1.116

Table 5: Slew-rate survival test. The reported row is the lowest voltage slew limit for which the controller remains within the nonlinear simulation constraints.

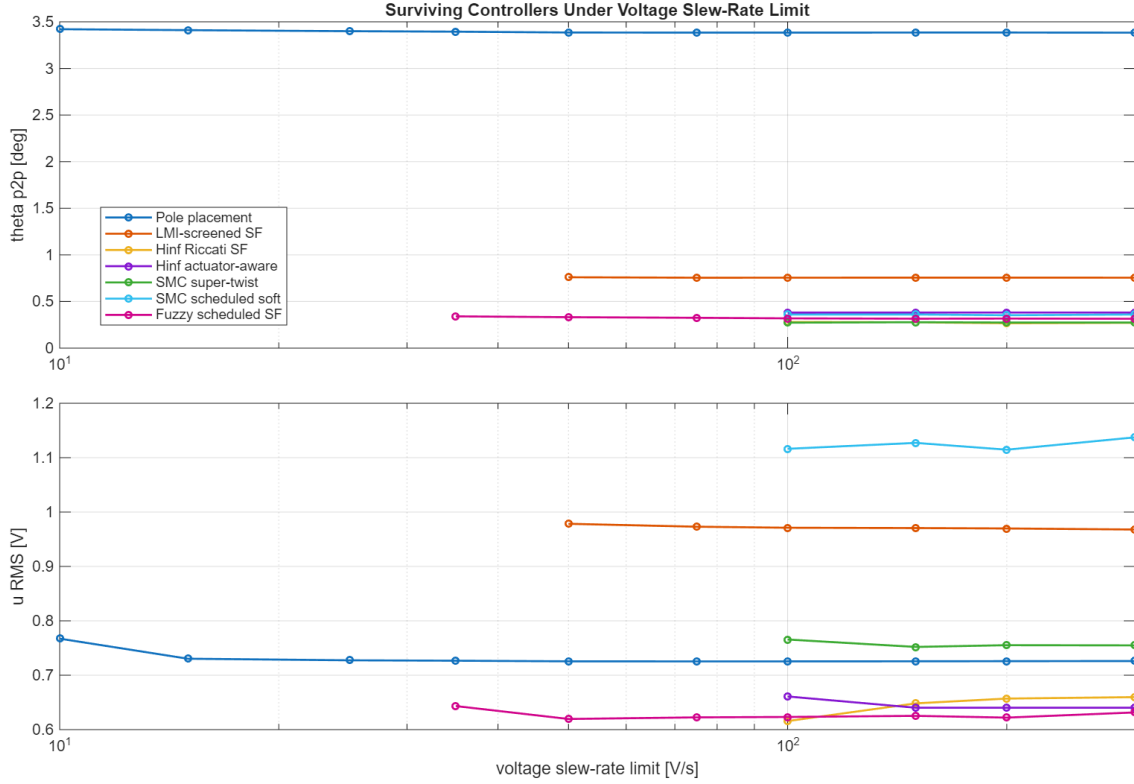


Figure 14: Controller survival under voltage slew-rate limits. Fuzzy scheduling is the best low-slew robust candidate: it remains below 0.35° peak-to-peak at 35 V/s, while the more aggressive H_∞ and SMC candidates require about 100 V/s.

The low-slew test changes the preferred controller. If the hardware can tolerate slew rates of approximately 100 V/s or higher, the H_∞ Riccati and SMC controllers give the smallest angle envelopes. If the design priority is weak-hardware protection, the fuzzy scheduled controller is more attractive: it is the best surviving robust controller at 75, 50, and 35 V/s, with $\theta_{pp} = 0.341^\circ$ at 35 V/s. Below that level only the pole-placement controller survives in this benchmark, but it returns to the original large limit cycle of approximately 3.4° peak-to-peak.

Based on these results, the most promising next hand-designed controller is the fuzzy scheduled state-feedback law. It does not give the absolute smallest nominal peak-to-peak angle, but it offers the best compromise between strong rocking suppression and low voltage slew-rate requirements. The next design step is therefore to replace the automatically swept fuzzy scheduler by an explicitly engineered rule base, using the robust-comparison results as the target envelope.

5. Experience 3: Periodic trajectory tracking for the body and lift-up

References

- [1] Chi-Tsong Chen. *Linear System Theory and Design*. Oxford University Press, 3 edition, 1999.

A. Appendix A

If you need to include an appendix to support the research in your thesis, you can place it at the end of the manuscript. An appendix contains supplementary material (figures, tables, data, codes, mathematical proofs, surveys, ...) which supplement the main results contained in the previous sections.

B. Appendix B

It may be necessary to include another appendix to better organize the presentation of supplementary material.